

RAWAN ALQAHTANI

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INFORMATION TECHNOLOGY | AI & DATA SCIENCE TRACK

SUMMARY

Information Technology graduate with an AI & Data Science track and hands-on experience in applied AI, machine learning, NLP, computer vision, data curation, and end-to-end system development. Experienced in independently owning AI systems and collaborating effectively across AI and research teams at KACST.

EXPERIENCE

AI & Software Development Intern | King Abdulaziz City for Science and Technology (KACST) Jun 2026 – Present

- Own end-to-end development of SafeBite, a bilingual AI-assisted food-safety system integrating Flutter/Firebase, FastAPI, barcode lookup, Google Cloud Vision OCR, deterministic ingredient rules, and Gemini fallback.
- Curated and processed 22K+ SFDA barcode records and benchmarked multilingual classification approaches on 80 manually labeled ingredient cases.
- Collaborated closely with the AI team to annotate and quality-check 200+ satellite images, aligning on labeling guidelines and resolving ambiguous cases to maintain dataset consistency.
- Collaborating with researchers on a KACST-MIT Jibo social robotics study, conducting SEL literature review and developing structured behavioral measures aligned with CASEL and SSIS SEL.

SELECTED PROJECTS

Wadiah | AI-Assisted Lost & Found System — Graduation Project Sep 2025 – May 2026

- Led development of an Arabic/English Flutter/Firebase lost-and-found system for Al-Masjid Al-Haram, covering reporting, staff verification, ownership review, secure PIN-based handover, and full lifecycle tracking.
- Built a multimodal retrieval pipeline using MobileNetV3Small/TFLite visual embeddings and multilingual MiniLM text embeddings, with cosine similarity, Top-5 candidate retrieval, dynamic indexing, staff-initiated rematching, and human-in-the-loop ownership confirmation.

Early Cardiovascular Disease Classification | Applied Machine Learning Aug 2025 – Dec 2025

- Cleaned and engineered features on ~70K patient records and compared Logistic Regression, SVM, and Random Forest using stratified cross-validation and GridSearchCV; tuned Random Forest achieved 73.7% accuracy and 0.802 ROC-AUC.

Emotion Classification on Weakly Labeled Humor Data | NLP Project Feb 2026 – May 2026

- Built TF-IDF-based Logistic Regression, Naive Bayes, and Linear SVM models on 200K humor samples, comparing text-only against weak-emotion features; best SVM achieved 94.95% accuracy and macro-F1.

Saudi PDPL Legal Text Analytics | Data Science & NLP Project Aug 2025 – Dec 2025

- Performed EDA and comparative analysis on 43 Saudi PDPL articles, interpreting article-length, TF-IDF, keyword-frequency, and modified-vs-original text patterns.
- Generated Q&A pairs per article using the Gemini LLM, comparing zero-shot, one-shot, and few-shot prompting strategies for Arabic legal text understanding.

RUH Flights Big Data Analytics | Apache Spark & Scala Project Feb 2026 – May 2026

- Independently implemented the Spark ML phase on 153K+ flight records, including feature preparation, Random Forest training, baseline comparison, model evaluation, and interpretation; also contributed to data cleaning and outlier handling.

EDUCATION

Bachelor of Information Technology | King Saud University

AI & Data Science Track

Sep 2022 – Aug 2026

SKILLS

Programming: Python, SQL, Java, JavaScript, Scala

AI & Data: Machine Learning, Deep Learning, NLP, Computer Vision, scikit-learn, TensorFlow, Pandas, NumPy, LLM workflows

Data & Analytics: Apache Spark, MySQL, Excel, Power BI, Data Cleaning, Data Annotation

Development & Tools: FastAPI, Flutter, Firebase, REST APIs, Git, GitHub, Jupyter, Google Colab, VS Code, Linux / Ubuntu

Languages: Arabic (Native), English (Advanced – C1)